Math 202: Differential Equations

Instructor and lectures:

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Assistant(s):

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Course Web page: Course announcements will be made on `feritozturk.github.io/here/f26m202.html`. You are responsible from all announcements made on this webpage or during the classes.

Problem Sessions: TBA. Please attend problem sessions regularly.

Prerequisites: (math 101 or math 131) and (math 201 or math 221)

Textbook:
Boyce&DiPrima, *Elementary Differential Equations & Boundary Value Prob.*

Grading:

- One midterm (4 questions): Nov 11, 2026; 17:00-19:00 (36%),
- Final exam (6 questions): t.b.a. (48%),
- 4 Pop-quizzes: each (4%) (5 quizzes; best 4 count).

Exam policies:

- In any of the exams above, you are forbidden to use any electronic device or to take help from any LLM. Any fraud during the exams or quizzes will be punished by an F grade! No discipline committees, no negotiations!
- To be eligible to take the final exam your total grade from the midterm exam and the quizzes must be at least 10 out of $52(= 16 + 36)$.
- Bring an official IDENTIFICATION CARD (Boğaziçi University Student ID Card, Nüfus Cüzdani, Drivers License or Passport) to the exams. No other form of ID is accepted.
- You can see your exam papers if you follow the classes regularly. Otherwise you may not see your exam papers.

Make-up Policy: A valid documented excuse is required to take a make-up exam for the midterm. Your document should be given to your instructor within a week after the exam. If you happen to miss the final exam, you must submit your document to the university-wide excuse exam committee.

No make-up for quizzes. If you miss a quiz, that one's over. No negotitation! You may take the quizzes in any Section (but each Quiz.No.k only once).

The make-up exams will be after the final, with the midterm and final make-ups at the same time. You will only be able to make up one exam in total. So if you miss the midterm and the final, you will only be able to make up one of them. If you miss the midterm, you can still take the final. However if you want to make up the midterm your final exam must be at least 10 out of 48.

Follow the announcements on the webpage for the make-up exam dates and the necessary instructions.

Students with disabilities:

- Students with special needs should see the instructor as soon as possible
- Additionally, they should contact the instructor at least 5 days prior to each exam.

Code of Conduct:

- We are here to **discuss** mathematics. Discussing is a 2-way process.

- We come here for **scientific exchange**. I try to put across a mathematical line of reasoning with you. Either you are convinced or you are not, in which case you may choose to object. Either I am convinced and I change my reasoning or I keep my claim, modify my expressions. This goes on back and forth.

- In any case we are here to **interact** with each other. Use this fantastic opportunity to advance your understanding in maths.

- If you attend the classes regularly, you will observe that you gain something more precious than earning points in an exam: Step by step you will enhance your intellectual capacity. Thus it is weird to skip classes.

- The process of seeing your exam papers is a continuation of this learning process. You see your papers to learn, not to **object** or to harrasse your assistants.
  If you attend more than 3/4 of the classes, you can see your exam papers.

- Whatever you tell me, whatever you write to me, whatever you submit to me, that must come from your intellect; it must be **your product**. You must understand what you are saying; you must write your own texts, with your own words and ideas. I strongly discourage you to use the output of LLMs unless you really understand the argument.

- If you use LLMs to produce work, you have already agreed to be **below average**. Any text that you get from an LLM is conventional.

- While you are in Boğaziçi, seize this opportunity to prosper your intellectual life.

- Needless to say any **grade optimization** is your play, not mine. Do not come to me with requests of letter grade modifications.

1. Introduction	1.1 Some Basic Mathematical Models; Direction Fields 1.2 Solutions of Some Differential Equations 1.3 Classification of Differential Equations
2. First Order Differential Equations	2.1 Linear Equations; Method of Integrating Factors 2.2 Separable Equations 2.4 Differences between Linear and Nonlinear Equations 2.6 Exact Equations and Integrating Factors 2.8 The Existence and Uniqueness Theorem
3. Second Order Linear Equations	3.1 Homogeneous Equations with Constant Coefficients 3.2 Solutions of Linear Homogeneous Equations; the Wronskian 3.3 Complex Roots of the Characteristic Equation 3.4 Repeated Roots; Reduction of Order 3.5 Non-homogeneous Eqns.; Method of Undetermined Coefficients 3.6 Variation of Parameters
4. Higher Order Linear Equations	4.1 General Theory of nth order Linear Equations 4.2 Homogeneous Equations with Constant Coefficients 4.3 The Method of Undetermined Coefficients 4.4 The Method of Variation of Parameters
5. Series Solutions of Second Order Linear Equations	5.1 Review of Power Series 5.2 Series Solutions Near an Ordinary Point, Part I 5.3 Series Solutions Near an Ordinary Point, Part II 5.4 Euler Equations; Regular Singular Points 5.5 Series Solutions Near a Regular Singular Point, Part I 5.6 Series Solutions Near a Regular Singular Point, Part II
6. The Laplace Transform	6.1 Definition of the Laplace Transform 6.2 Solution of Initial Value Problems 6.3 Step Functions 6.4 Differential Equations with Discontinuous Forcing Functions 6.5 Impulse Functions 6.6 The Convolution Integral
7. Systems of First Order Linear Equations	7.1 Introduction 7.4 Basic Theory of Systems of First Order Linear Equations 7.5 Homogeneous Linear Systems with Constant Coefficients 7.6 Complex Eigenvalues 7.7 Fundamental Matrices 7.8 Repeated Eigenvalues 7.9 Non-homogeneous Linear Systems
10. Partial Differential Equations and Fourier Series	10.1 Two-Point Boundary Value Problems 10.2 Fourier Series 10.3 The Fourier Convergence Theorem 10.4 Even and Odd Functions 10.5 Separation of Variables; Heat Conduction in a Rod 10.6 Other Heat Conduction Problems 10.7 The Wave Equation: Vibrations of an Elastic String